

Neurons and glia

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The five worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 7 TO 11 OF WEEK 4

23 SLIDES · 5 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

A neuron has one job, which is to carry a signal from the place where it receives information to the place where it passes that information on. This lecture looks at how the cell is built to do that, and at the other cells in nervous tissue, the glia, that keep it working. It is a physiology lecture, so for every structure the question is the same: what does this part do to move a signal from input to output?

Here is the problem we solve by the end. A 28 year old graduate student wakes up with blurred vision in her right eye, and the eye hurts when she moves it. Over several weeks her vision comes back. Months later she notices that in a hot shower, or on a run on a hot day, the vision in that eye blurs again, and within an hour of cooling down it clears. Her eye is healthy, and every neuron that carries the signal from that eye to her brain is still there.

Her results at the clinic, on a cool day, are these. Plasma potassium is 4.1 mM (range 3.5 to 5.0) and plasma sodium is 140 mM (range 135 to 145), so the resting potential you built in the last lecture is normal. Her temperature at rest is 36.9 °C (98.4 °F). The informative result is a timing test called a visual evoked potential. She watches a checkerboard pattern that flips back and forth, one eye at a time, and electrodes on the scalp over the visual cortex record how long the signal takes to arrive. Normal is about 100 ms, up to about 115 ms, and the limit varies a little between laboratories. From her left eye the signal takes 102 ms. From her right eye it takes 142 ms. The signal arrives, and it is close to its normal size, but it is about 40 ms late. So the questions are what sets the speed of a signal along an axon, what could change that speed in one optic nerve and not the other, and why warming her body would make it worse.

See it move: send cargo along an axon

SLIDE 19 OF THE DECK

This is a working model of axonal transport, which is covered in the second half of these notes. It is placed here so that you can find it easily. Choose a cargo, 1 to 4, and watch which motor protein carries it, which way it goes and how far it has traveled. The table beside the drawing converts what you see into real time, using typical textbook speeds. You can change the length of the axon with the slider, from 1 mm for a short interneuron in the brain to 1 m for a motor neuron that reaches the foot, and you can remove ATP. The animation is sped up enormously and is not to scale, so read the real times from the table.

Moving model: run it on the course site, drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-neurons-glia-notes.html

THREE THINGS TO TRY

- 1 Set the axon length to 1 m. Send cargo 1, the synaptic vesicle protein, and then cargo 2, the cytoskeleton protein. Write down the real time for each whole trip and compare them with worked problem 4.
- 2 Choose cargo 4, the virus. Notice that it starts at the axon terminal and that the motor is dynein. Then slide the axon length from 1 m down to 100 mm and see how the time to reach the cell body changes. That is the difference between a bite on the foot and a bite much closer to the spinal cord.
- 3 While any cargo is moving, press the ATP button to remove ATP. Every motor stops. Put the ATP back and they walk again.

Two kinds of cells, and neurons sorted by shape

SLIDES 3 TO 4 OF THE DECK

Nervous tissue has two kinds of cells. Neurons carry electrical signals. Glia, also called glial cells or neuroglia, keep the neurons working, and most of them do not carry signals. The word glia comes from the Greek word for glue, because early anatomists thought these cells only held the tissue together. Two abbreviations are used all through this unit. The central nervous system, or CNS, is the brain and the spinal cord. The peripheral nervous system, or PNS, is every nerve and every ganglion outside them. A ganglion is a cluster of neuron cell bodies outside the CNS.

A neuron has a cell body, which holds the nucleus and makes the cell's proteins, and it has processes, which are the extensions that leave the cell body. Dendrites are processes that receive signals. The axon is the process that carries the signal away from the cell body to the axon terminal. The structural classes are named by counting the processes that leave the cell body.

The functional classes describe the direction in which a neuron carries its signal. Sensory neurons, also called afferent neurons, carry signals from receptors toward the CNS. Interneurons lie entirely inside the CNS and connect neurons to each other. Motor neurons, also called efferent neurons, carry signals away from the CNS to muscles and glands. Afferent means carrying toward, and efferent means carrying away. The efferent group also includes the autonomic neurons that control smooth muscle, heart muscle and glands, which come later in the course.

Shape and job are two separate labels, but they pair up in a regular way. Most sensory neurons are pseudounipolar, and the sensory neurons for smell, vision and hearing are bipolar. Interneurons and motor neurons are multipolar, and a few interneurons are anaxonic. Interneurons outnumber the other two classes enormously. A common textbook estimate is that about 99 of every 100 neurons are interneurons, which tells you that most of what the CNS does is process information between the input and the output.

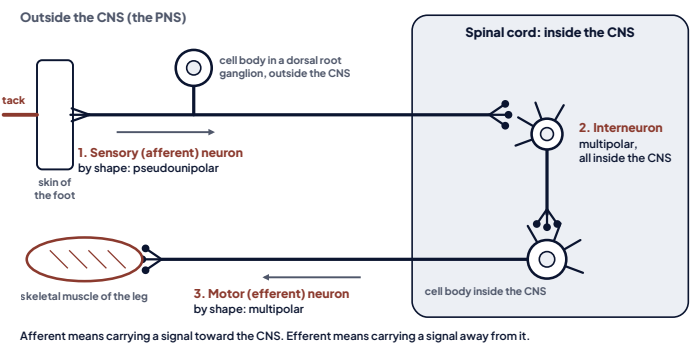


Figure 2. You step on a tack and your leg pulls back. Three neurons in a row carry the signal, and each one is labeled by job and by shape.

The pseudounipolar shape has a functional advantage. With the cell body off to the side, the signal can travel from the receptor in the skin to the spinal cord along one continuous axon, and it never has to cross the cell body. In a multipolar neuron, signals from the dendrites have to spread across the cell body before a new action potential can start, and signals that spread that way fade with distance. A touch neuron from the toe is about a meter long, and its action potential starts near the receptor and runs the whole way without interruption. The cell body still does its other job, which is to make the proteins that keep the cell alive.

Signals travel from the top of each drawing to the bottom

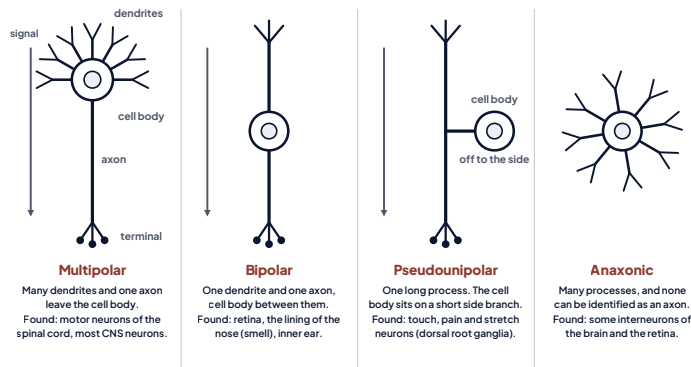


Figure 1. The four structural classes, with the signal traveling from the top of each drawing to the bottom.

Process

Any extension that leaves the cell body of a neuron: a dendrite or an axon.

Dendrite

A process that receives incoming signals.

Axon

The single process that carries the signal away from the cell body to the axon terminal.

Ganglion

A cluster of neuron cell bodies outside the CNS.

Multipolar, bipolar, pseudounipolar, anaxonic

Many dendrites and one axon; one dendrite and one axon; one fused process with the cell body on a side branch; many processes with no identifiable axon.

Neurons sorted by job

WORKED PROBLEM 1

You step on a tack and your leg pulls back. Label each of the three neurons in the pathway twice, once by shape and once by job, and say whether its cell body is inside or outside the CNS.

Use Figure 2 with the labels covered.

THE WORK

- 1** **Neuron 1** runs from the skin to the spinal cord, so it carries the signal toward the CNS. By job it is a **sensory (afferent)** neuron. By shape it is **pseudounipolar**. Its cell body is in a dorsal root ganglion, which is **outside** the CNS.
- 2** **Neuron 2** begins and ends inside the spinal cord. By job it is an **interneuron**. By shape it is **multipolar**. Its cell body is **inside** the CNS, as is every other part of it.
- 3** **Neuron 3** runs from the spinal cord to the muscle, so it carries the signal away from the CNS. By job it is a **motor (efferent)** neuron. By shape it is **multipolar**. Its cell body is **inside** the CNS, in the ventral horn of the spinal cord, and only its axon leaves.
- 4** Check the pattern. Shape and job are two separate labels, but they pair up in a regular way. Most sensory neurons are pseudounipolar, and the sensory neurons for smell, vision and hearing are bipolar. Interneurons and motor neurons are multipolar, and a few interneurons are anaxonic.
- 5** Interneurons outnumber the other two classes enormously. A common textbook estimate is that about 99 of every 100 neurons are interneurons. That tells you most of what the CNS does is process information between the input and the output.

Sensory (afferent) neuron

Carries signals from a receptor toward the CNS. Its cell body is in a ganglion outside the CNS.

Interneuron

A neuron that lies entirely inside the CNS and connects other neurons. By far the most numerous class.

Motor (efferent) neuron

Carries signals away from the CNS to a muscle or gland. Its cell body is inside the CNS.

Two kinds of electrical signal are involved, and the next lecture explains how both are produced. A graded potential is a small change in membrane potential whose size depends on the strength of the stimulus, and it fades as it spreads. An action potential is always the same size and does not fade, which is what all or none means. For now you need to know where in the cell each one happens, and the answer follows from which ion channels are in the membrane of each region. These are the four channel types from the last lecture.

The input zone has ligand gated channels, which open when a neurotransmitter binds, and the receptor endings of sensory neurons have mechanically gated channels and other channels that respond to a physical stimulus. These channels produce graded potentials. The trigger zone has voltage gated Na^+ channels at a higher density than anywhere else in the cell. If the graded potential that reaches it is still at threshold, about -55 mV , those channels open and an action potential begins. If the graded potential has faded below threshold, nothing is sent. The axon has voltage gated Na^+ and K^+ channels, which make the action potential again and again along its length. The axon terminal has voltage gated Ca^{2+} channels. Free Ca^{2+} is about 1 mM outside the cell and 0.0001 mM inside, and E_{Ca} is about $+123 \text{ mV}$, so when the action potential opens these channels, Ca^{2+} enters. The rise in Ca^{2+} inside the terminal makes vesicles fuse with the membrane and release neurotransmitter, and the electrical signal becomes a chemical one.

If you shade the regions where graded potentials happen, then the regions where voltage gated Na^+ channels are dense, and then the region where the signal becomes chemical, only one region is shaded more than once. Graded potentials spread as far as the trigger zone, and the trigger zone has the densest voltage gated Na^+ channels. That overlap is what makes it the decision point of the neuron. One detail is different in a sensory neuron: its trigger zone is not at the cell body. It is at the far end, where the receptor endings join the axon.

You can test your map of the neuron by blocking one kind of channel at a time. Tetrodotoxin, from pufferfish, blocks voltage gated Na^+ channels. The input zone still produces graded potentials, but no action potential can start at the trigger zone. Omega-conotoxin, from cone snail venom, blocks the voltage gated Ca^{2+} channels of axon terminals. Action potentials arrive at the terminal normally, but no Ca^{2+} enters and no neurotransmitter is released. Curare, a plant arrow poison, blocks the ligand gated channels on skeletal muscle, so the whole neuron works and the signal stops at the input zone of the next cell. The muscle fails to contract in all three cases, and to say which region failed you have to know which channel is found there.

One neuron, four working regions

SLIDES 6 TO 8 OF THE DECK

A neuron can be divided into four regions by what each one does. The input zone is the dendrites and the cell body. It receives signals. The trigger zone is the axon hillock, where the axon leaves the cell body, together with the first stretch of the axon, called the initial segment. It decides whether a signal is sent. The conducting zone is the axon, which carries the signal over a distance. The output zone is the axon terminal, which passes the signal to the next cell.

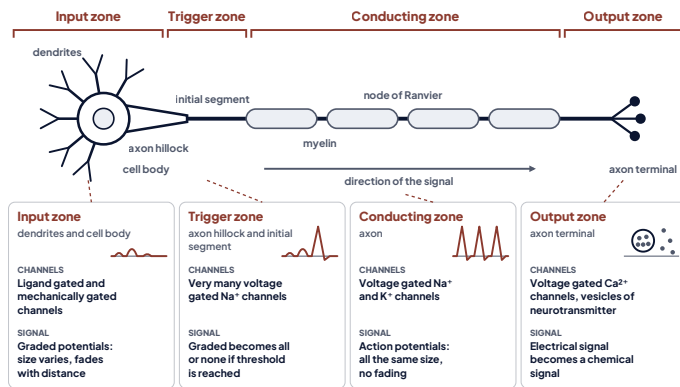


Figure 3. A motor neuron read from left to right, with the channels and the signal type for each region.

Input zone

Dendrites and cell body. Ligand gated and mechanically gated channels. Graded potentials.

Trigger zone

Axon hillock and initial segment. The highest density of voltage gated Na^+ channels. The place where a graded potential becomes an action potential if threshold is reached.

Conducting zone

The axon. Voltage gated Na^+ and K^+ channels. Action potentials.

Output zone

The axon terminal. Voltage gated Ca^{2+} channels and vesicles of neurotransmitter. The electrical signal becomes a chemical signal.

Graded potential and action potential

A graded potential varies in size and fades with distance. An action potential is all or none: always the same size, with no fading.

The glia of the CNS

SLIDES 9 TO 10 OF THE DECK

The CNS has four kinds of glial cell. Astrocytes are star shaped cells that control the fluid around neurons. Oligodendrocytes make the myelin of the CNS, and one oligodendrocyte wraps segments on several axons. Microglia are the immune cells of the CNS, and they engulf debris, dead cells and microbes.

Ependymal cells form the lining of the fluid filled spaces of the brain and spinal cord. Their cilia move the cerebrospinal fluid, and specialized ependymal cells in the choroid plexus produce it.

Astrocytes do four things that matter for signaling. First, they take up K^+ . Every action potential lets a little K^+ out of the neuron, and the space between cells in the brain is very narrow, so that K^+ would build up quickly. Astrocytes take it up and pass it through a network of connected astrocytes to where the concentration is lower. Second, they take up neurotransmitter. Astrocytes wrap around synapses and remove glutamate and other neurotransmitters, which ends the signal, and they return the material to the neuron to be used again. Third, they pass fuel to neurons. They take glucose from the blood and supply neurons with lactate, which the neurons use to make ATP.

Fourth, astrocytes help form the blood brain barrier. The barrier itself is the tight junctions between the cells of the capillary wall,

which seal brain capillaries far more tightly than capillaries elsewhere. Astrocyte end feet cover the capillaries and signal those cells to form the tight junctions and keep them. Small lipid soluble molecules such as O_2 , CO_2 and alcohol cross freely. Glucose and amino acids cross on transport proteins. Many drugs cannot cross, which is why dopamine cannot be given for Parkinson disease and levodopa, which rides on an amino acid transporter, is given instead.

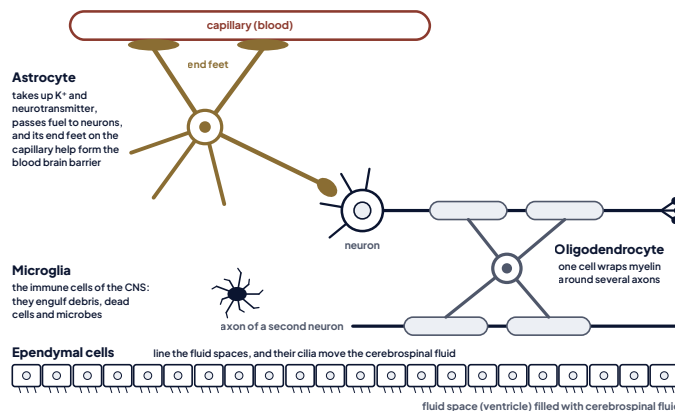


Figure 4. The four glial cells of the CNS, each drawn doing its job.

WORKED PROBLEM 2

The astrocytes around a busy group of neurons stop clearing K^+ , and the K^+ outside those neurons rises from 4 mM to 8 mM. Find the new E_K and the new resting potential.

Keep 140 mM K^+ inside, Na^+ at 145 out and 15 in, and $P_{\text{Na}} = 0.04$, as in the last lecture.

THE WORK

- Potassium equilibrium potential with 8 mM outside:
 $E_K = 61.5 \times \log_{10}(8 \div 140) = 61.5 \times (-1.243) = -76 \text{ mV}$.
 With the normal 4 mM it was -95 mV .
- Goldman equation, with $P_K = 1$ and $P_{\text{Na}} = 0.04$ as in the last lecture. Top: $8 + (0.04 \times 145) = 8 + 5.8 = 13.8$. Bottom: $140 + (0.04 \times 15) = 140.6$.
- Ratio: $13.8 \div 140.6 = 0.0981$. $\log_{10}(0.0981) = -1.008$.
 $V_m = 61.5 \times (-1.008) = -62 \text{ mV}$.
- The resting potential of every neuron in that region has moved from -71 mV to -62 mV . Threshold is about -55 mV , so these neurons are now only 7 mV away from firing, and they were never stimulated.
- This is what astrocytes prevent. Every action potential lets a little K^+ out of the neuron, and the space between cells in the brain is very narrow, so the K^+ concentration there can rise quickly. Astrocytes take the K^+ up and pass it through the network of astrocytes to places where the concentration is lower. When that clearing fails, activity in one group of neurons can spread to its neighbors, which is one of the ways a seizure spreads. The 8 mM here is an illustration and not a measured value for a particular condition.

Astrocyte

CNS glial cell that takes up K^+ and neurotransmitter, passes fuel to neurons and helps form the blood brain barrier.

Oligodendrocyte

CNS glial cell that makes myelin. One cell wraps segments on several axons.

Microglia

The immune cells of the CNS. They engulf debris, dead cells and microbes.

Ependymal cell

Lines the fluid filled spaces of the brain and spinal cord, moves the cerebrospinal fluid with its cilia, and in the choroid plexus produces it.

Blood brain barrier

Tight junctions between the cells of brain capillary walls, formed and kept under the influence of astrocyte end feet, that limit what passes from blood into brain tissue.

The glia of the PNS, and what myelin is

SLIDE 11 OF THE DECK

The PNS has two kinds of glial cell. Satellite cells surround neuron cell bodies in ganglia. They support the cell body and control the fluid around it, much as astrocytes do in the CNS. Schwann cells make the myelin of the PNS. Each Schwann cell wraps one segment, about 1 mm long, of one axon, so a long axon needs hundreds of Schwann cells in a row. The gaps of bare axon membrane between segments are the nodes of Ranvier.

Myelin is the glial cell's own membrane wound around the axon many times, up to about 150 layers in a large axon. Membrane is mostly lipid, and ions cannot cross lipid, so myelin is electrical insulation. Two of the six glial cells make myelin, and they differ in one structural way. An oligodendrocyte sends out many arms, each arm wraps one segment, and one oligodendrocyte can supply segments to several dozen axons. A Schwann cell wraps its whole body around a single segment of a single axon. This matters in disease, because losing one oligodendrocyte strips a segment from many axons at once.

In a ganglion and a nerve of the PNS

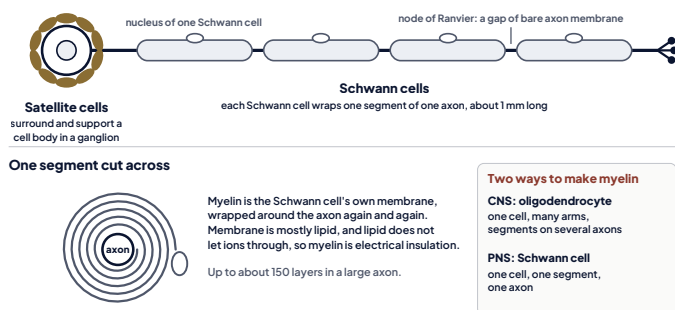


Figure 5. Satellite cells and Schwann cells, with one myelin segment cut across to show its layers.

Schwann cell

PNS glial cell that makes myelin. One cell wraps one segment of one axon.

Satellite cell

PNS glial cell that surrounds and supports a neuron cell body in a ganglion.

Myelin

Many layers of glial cell membrane wound around an axon. It is electrical insulation.

Node of Ranvier

A short gap of bare axon membrane between two myelin segments, packed with voltage gated Na^+ channels.

How myelin speeds conduction, and what happens when it is lost

SLIDES 12 TO 15 OF THE DECK

An action potential does not slide along the axon. Each patch of membrane has to be depolarized to threshold by current from the patch before it, and then it has to open its own Na^+ channels. Opening channels is the slow step. Current flowing along the inside of the axon is fast, but in a bare axon it leaks out through the membrane, so it reaches only a short distance ahead, and the action potential has to be made again at every patch. An unmyelinated axon conducts at about 0.5 to 2 meters per second.

Myelin carries no signal itself. It stops the leak. Under the myelin, current from one node flows along the inside of the axon and reaches the next node, about 1 mm away, still strong enough to bring that node to threshold. The action potential is made again only at the nodes, and this node to node pattern is called saltatory conduction. A large myelinated axon conducts at up to about 120 meters per second. A wider axon also conducts faster, because a wide axon has less resistance to current flowing along it, and myelin gives the same speed in a far thinner axon. The full electrical explanation comes with the action potential lectures.

Voltage gated Na^+ channels are not spread evenly along a myelinated axon. They are packed densely at the nodes, and there are very few under the myelin. When a segment of myelin is stripped away, two things go wrong together. Current leaks out through the bare membrane, and the bare membrane has too few Na^+ channels to make a new action potential. The depolarization fades before it reaches the next node, that node stays below threshold, and the signal stops. So conduction in a demyelinated axon does not drop to the speed of an unmyelinated axon. It fails, and this is called conduction block. If the myelin is only thinned, the signal gets through, but late. Over weeks the axon can add Na^+ channels to the bare stretch, and the glial cells can partly remyelinate it, so some function returns.

Two diseases show the difference between CNS and PNS myelin. In multiple sclerosis the immune system attacks the myelin of the CNS, which is made by oligodendrocytes, in the brain, the spinal cord and the optic nerves. What the person notices depends on which pathway is hit, such as blurred vision, numbness, weakness or poor balance, and episodes come and go over years. Recovery is partial, because oligodendrocytes remyelinate poorly. In Guillain-Barré syndrome the immune system attacks the myelin of the PNS, which is made by Schwann cells, often a few weeks after an infection. Weakness and tingling start in the feet and hands and move upward over days, and reflexes are lost. Most people recover over weeks to months, because Schwann cells remyelinate well.

Two axons, each 1 meter long. A dot marks every place the action potential has to be made again.

Unmyelinated axon



Myelinated axon

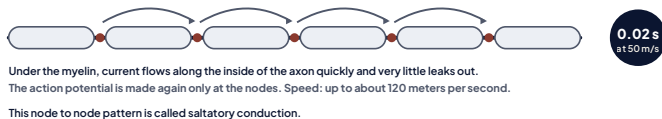


Figure 6. A dot marks every place the action potential has to be made again. Myelin reduces the number of those places.

WORKED PROBLEM 3

You stub your toe. A touch signal travels in a large myelinated axon at about 50 m/s, and the dull ache travels in a thin unmyelinated axon at about 1 m/s. The toe is about 1 m from the spinal cord. How long does each signal take to arrive?

time = distance ÷ speed.

THE WORK

- 1 Use time = distance ÷ speed, and keep the units matched: meters and meters per second.
- 2 Touch, in a large myelinated axon: $1 \text{ m} \div 50 \text{ m/s} = 0.02 \text{ s}$, which is 20 milliseconds.
- 3 Dull ache, in a thin unmyelinated axon: $1 \text{ m} \div 1 \text{ m/s} = 1 \text{ s}$.
- 4 The touch signal arrives 50 times sooner. You have felt this. When you stub your toe you know at once that you hit something, and the ache arrives about a second later. That delay is the difference between a myelinated and an unmyelinated axon.
- 5 The sharp first pain that comes before the ache travels in thinly myelinated axons at about 5 to 30 m/s, which puts it between the two. All of these speeds are approximate, and books give ranges.

Intact myelin



Middle segment stripped bare

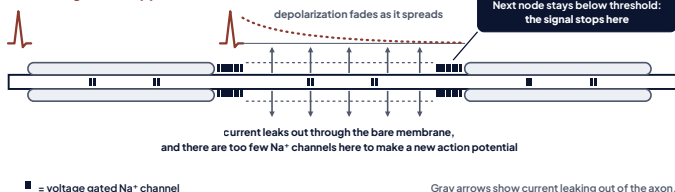


Figure 7. The same axon with its myelin intact, and with the middle segment stripped.

Saltatory conduction

Conduction in a myelinated axon, in which the action potential is regenerated only at the nodes of Ranvier.

Demyelination

Loss of myelin from an axon.

Conduction block

Failure of an action potential to pass a damaged stretch of axon.

Multiple sclerosis

A disease in which the immune system attacks CNS myelin, made by oligodendrocytes.

Guillain-Barré syndrome

A disease in which the immune system attacks PNS myelin, made by Schwann cells.

Axonal transport

SLIDES 16 TO 20 OF THE DECK

Almost every protein in a neuron is made in the cell body, because that is where the nucleus, the rough ER and the Golgi are. The axon and its terminal have almost no ribosomes. Yet the terminal needs a steady supply of channels and pumps for its membrane, the proteins of synaptic vesicles, enzymes that make neurotransmitter, and mitochondria, and proteins wear out in days to weeks. The distances are large. The cell body of a large motor neuron is about 0.1 mm across, and its axon to the foot is about 1 m long, which is 10,000 times the width of the cell body. If the cell body were the size of a tennis ball, the axon would be a tube about 670 m long, which is about 0.4 mile. Diffusion cannot cover that distance, because the time diffusion takes rises with the square of the distance. A protein can diffuse across a cell body in seconds, but to diffuse 1 m would take hundreds of years at the least. A neuron therefore needs a transport system that carries cargo in one direction, along a track, using energy.

The tracks are microtubules, which run the length of the axon. Every microtubule in an axon points the same way, with its plus end toward the terminal and its minus end toward the cell body. Two motor proteins walk along them, and each step splits ATP. Kinesin walks toward the plus end, so it carries cargo anterograde, from the cell body to the terminal. Dynein walks toward the minus end, so it carries cargo retrograde, from the terminal back to the cell body.

Fast anterograde transport carries vesicles, membrane proteins such as channels, and mitochondria at about 400 mm per day. Fast retrograde transport is somewhat slower, about 200 to 300 mm per day. It brings worn out membrane and old organelles back to the cell body, where lysosomes break them down and the parts are reused. It also carries information: growth factors released by the target cell are taken up at the terminal and carried back to the nucleus, and they tell the neuron that its connection is alive. Slow transport moves cytoskeleton proteins and dissolved enzymes from the cell body toward the terminal at about 0.2 to 8 mm per day. Current evidence is that this cargo rides on the same motors but spends most of its time paused, so its average speed is low. Some books describe slow transport as a flow of the cytoplasm, which was the earlier explanation. All of these speeds are approximate.

Some viruses and toxins reach the cell body by riding on dynein. Rabies virus enters the axon terminals of motor neurons near a bite, travels to the spinal cord, and spreads from neuron to neuron to the brain. A bite on the foot gives the longest path, which is one reason the illness usually takes longer to appear after a bite on the

leg than after a bite on the face. Herpes simplex virus enters sensory axon terminals in the skin of the lip and is carried to the cell body in a sensory ganglion, where it stays dormant. When it reactivates, new virus travels back down the same axon on kinesin, so the cold sore returns in the same place. Tetanus toxin enters motor axon terminals near a dirty wound and is carried to the spinal cord. There it crosses to inhibitory interneurons and blocks their release of neurotransmitter, so muscles cannot relax.

Now change the conditions. If ATP runs out, every motor stops, fast and slow, in both directions. If a drug takes the microtubules apart, the motors have nothing to walk on, and the terminals farthest from their cell bodies run short first. Those belong to the longest axons, so the person notices numbness and tingling that begin in the feet and the hands. This happens with vincristine, a chemotherapy drug that blocks the assembly of microtubules. It is given to stop cancer cells from dividing, and the effect on long axons is a side effect.

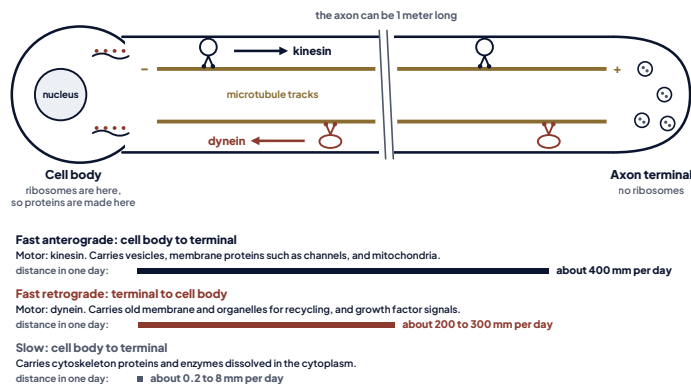


Figure 8. Three transport streams in one axon. The bars compare the distance each one covers in a day.

WORKED PROBLEM 4

A motor neuron has an axon 1 m long. How long does a protein take to reach the terminal by fast anterograde transport at 400 mm per day, and by slow transport at 1 mm per day? How long does the return take by fast retrograde transport at 200 mm per day?

time = distance ÷ speed, with 1 m = 1000 mm.

THE WORK

- 1** Use time = distance ÷ speed, with the distance in millimeters: 1 m = 1000 mm.
- 2** Fast anterograde transport: 1000 mm ÷ 400 mm per day = **2.5 days**. A sodium channel made in the cell body on Monday morning is in the membrane near the foot by Wednesday evening.
- 3** Slow transport at 1 mm per day: 1000 mm ÷ 1 mm per day = 1000 days, and 1000 ÷ 365 = **about 2.7 years**.
- 4** Slow transport covers a range. At 8 mm per day the same distance takes 125 days. At 0.2 mm per day it takes 5000 days, which is more than 13 years. The cytoskeleton proteins are the slowest cargo.
- 5** Fast retrograde transport at 200 mm per day: 1000 ÷ 200 = **5 days** from the terminal back to the cell body.
- 6** What the numbers say: fast transport can resupply the far end of the longest axon in the body within a few days, which is about how often membrane proteins need replacing. Slow transport is several hundred times slower, and that matters when an axon has to regrow, as you will see on slide 21.

Axonal transport

Movement of material along an axon by motor proteins walking on microtubules, using ATP.

Anterograde and retrograde

Anterograde is from the cell body toward the terminal. Retrograde is from the terminal toward the cell body.

Kinesin

The motor protein for anterograde transport. It walks toward the plus end of the microtubule.

Dynein

The motor protein for retrograde transport. It walks toward the minus end of the microtubule.

Fast and slow transport

Fast: about 400 mm per day anterograde and 200 to 300 retrograde, for vesicles, membrane proteins and organelles. Slow: about 0.2 to 8 mm per day, for cytoskeleton proteins and dissolved enzymes.

When an axon is cut

SLIDE 21 OF THE DECK

When an axon is cut, the part that is separated from the cell body degenerates, because it no longer receives anything by axonal

transport. What happens next depends on the glial cells. In the PNS, Schwann cells survive. They help clear the debris, line up as a tube and release growth factors, and the stump of the axon can regrow along that tube at about 1 to 2 mm per day. That is about the speed of slow axonal transport, and it is not a coincidence, because a growing axon is built from cytoskeleton proteins that arrive by slow transport. In the CNS, oligodendrocytes and astrocytes release molecules that inhibit growth, and astrocytes form a scar. There is very little regrowth, so damage to the brain or spinal cord is largely permanent.

WORKED PROBLEM 5

A nerve is cut at the wrist and the two ends are stitched together. The fingertip is 200 mm from the cut. If the axons regrow at 1 to 2 mm per day, how long before feeling could begin to return in the fingertip? How long does an action potential at 50 m/s take to cover the same 200 mm?

time = distance ÷ speed.

THE WORK

- 1 The axon has to regrow from the cut to the fingertip, 200 mm.
- 2 At 2 mm per day: $200 \div 2 = 100$ days. At 1 mm per day: $200 \div 1 = 200$ days.
- 3 So feeling could begin to return in the fingertip after roughly 3 to 7 months, and there is also a delay of days to weeks before regrowth starts. These numbers are approximate.
- 4 Compare that with signaling. Once the axon is rebuilt, an action potential covers the same 200 mm at 50 m/s in $0.2 \div 50 = 0.004$ s, which is 4 milliseconds.
- 5 The regrowth speed is about the same as slow axonal transport, and that is not a coincidence. A growing axon is built from cytoskeleton proteins, and those arrive from the cell body by slow transport.

The patient, explained

SLIDE 22 OF THE DECK

The optic nerve is part of the CNS, so its axons are myelinated by oligodendrocytes. In multiple sclerosis the immune system attacks CNS myelin. Her first episode was a patch of inflammation in the right optic nerve that stripped myelin from many axons. Those axons went into conduction block, and her vision blurred. Then the inflammation settled, oligodendrocytes partly remyelinated the patch, and the axons added Na^+ channels to the bare membrane. Signals get through again, but slowly, which is the 142 ms against 102 ms in her timing test.

A repaired axon has very little to spare. In a healthy myelinated axon, each node receives several times more current than it needs to reach threshold. In her repaired axons the current that arrives is only just enough. Ion channels open and close faster when they are warmer, so in a warm axon the Na^+ channels inactivate sooner and each action potential delivers less current to the next node. A healthy axon has plenty to spare, so nothing changes. In her right optic nerve, a rise in body temperature of about 0.5°C (0.9°F) is

enough to drop the current below what the next node needs, and those axons stop conducting until she cools down. No new damage has been done. This is called the Uhthoff phenomenon.

Look at it as an experiment. The variable that fails is the current reaching the next node, compared with the current that node needs to reach threshold. Anything that makes each action potential last longer delivers more current. Lowering the temperature does that, because cooler channels inactivate later, and this is why her vision clears when she cools down. Blocking voltage gated K^+ channels does it too. Losing myelin uncovers K^+ channels that are normally hidden beneath it, and when they open, positive charge leaves the axon and the action potential is cut short. A drug that blocks these channels lengthens the action potential and improves conduction in some people with multiple sclerosis. Neither change repairs the myelin. They act on the electrical problem, and the immune attack is a separate target.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

What an ion is, and the charges on Na^+ , K^+ and Ca^{2+}
Lipids do not let ions through, which is why a membrane is an electrical insulator
ATP is the molecule cells split to pay for work

Chemistry review page

FROM EARLIER LECTURES

The resting potential, -70 mV, and threshold, about -55 mV (last lecture)
The Nernst and Goldman equations, and the typical ion concentrations (last lecture)
The four channel types by what opens them, and the words depolarize and hyperpolarize (last lecture)
Rough ER, Golgi, vesicles, lysosomes and the cytoskeleton (Week 2)
Diffusion is fast over short distances and slow over long ones (Week 3)

The only anatomy you need is this: the CNS is the brain and spinal cord, the PNS is everything else, and a ganglion is a cluster of neuron cell bodies outside the CNS.

MATH

time = distance ÷ speed
Unit conversions: $1\text{ m} = 1000\text{ mm}$, and $1\text{ s} = 1000\text{ ms}$
Converting days to years by dividing by 365

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open. Most of these prompts ask for a drawing, so practice the drawings from the figures above.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

07

Neuron structural and functional classes

Map the organization of the nervous system into the central nervous system, the sensory and efferent divisions of the peripheral nervous system, the somatic motor and autonomic branches, and the enteric nervous system, then classify a neuron as multipolar, bipolar, or pseudounipolar by structure and as sensory, motor, or interneuron by function, and match each class to a location in the nervous system.

Do one of the two. Head your paper with the number.

7-1

Draw a multipolar, a bipolar, and a pseudounipolar neuron side by side, each with its dendrites, cell body, and axon in the right arrangement and the cell body in the right position relative to the signal path. Under each write one real location in the body where you would find it. One line: one of the three has its cell body off to the side of the signal path, so say what advantage that gives a sensory neuron.

7-2

Draw a reflex arc across your box, from a receptor in the skin to a skeletal muscle, with three neurons in it. Label every neuron twice, once by structure and once by function, and mark whether each cell body sits inside or outside the CNS. One line: name the functional class that outnumbers the other two enormously in a human, and say what that tells you about what the CNS mostly does.

08

Functional regions of a neuron

Label the dendrites, cell body, axon hillock, trigger zone, axon, and axon terminal and state which signal type each region carries.

Do one of the two. Head your paper with the number.

8-1

Draw one motor neuron big, across the whole width of your box. Label the dendrites, cell body, axon hillock, trigger zone, axon, myelin, nodes, and axon terminal. Above each region write the signal type it carries, graded or all or none, and sketch the shape of that signal there. One line: mark the spot where graded becomes all or none and say what has to be true at that spot for the change to happen.

8-2

Draw the same neuron three times as three narrow strips stacked up. In the first, shade every region where graded potentials happen. In the second, shade every region with a high density of voltage gated sodium channels. In the third, shade where chemical signaling takes over from electrical. One line: name the region that appears in more than one strip and say why that makes it the decision point.

09

Glial cell functions

Match astrocytes, oligodendrocytes, microglia, ependymal cells, satellite cells, and Schwann cells to their physiological roles.

Do one of the two. Head your paper with the number.

9-1

Draw a CNS scene and a PNS scene side by side. In the CNS scene put an astrocyte touching both a capillary and a neuron, an oligodendrocyte myelinating more than one axon, a microglial cell, and ependymal cells lining a fluid space. In the PNS scene put a Schwann cell on one axon segment and satellite cells wrapped around a cell body. One line: two of these six make myelin, so say the one structural difference in how they do it.

9-2

Draw one neuron in the middle of your box with six arrows pointing at it, one from each glial type. On each arrow write the single job that glia does for the neuron or its surroundings. Then cross out one arrow of your own choosing and draw what goes wrong. One line: name the glial cell you removed and the first symptom you would expect, not the last.

10

Myelin and its loss

Explain how myelin and the nodes of Ranvier speed conduction and predict the functional consequence of demyelination.

Do one of the two. Head your paper with the number.

10-1

Draw two axons the same length side by side, one bare and one myelinated with nodes of Ranvier. On each, mark with dots every place the action potential has to be regenerated. Put a clock reading beside each showing which signal arrives first. One line: myelin is an insulator and does nothing electrical itself, so say in one sentence how adding insulation makes a signal faster.

10-2

Draw a myelinated axon with three intact internodes, then draw the same axon with the middle internode stripped bare. Show the action potential traveling along it and mark where it slows or fails. On both, draw the sodium channel distribution, dense at the nodes and sparse under the myelin. One line: the demyelinated stretch has few sodium channels, so say why conduction fails there rather than simply slowing down.

11

Axonal transport

Distinguish fast anterograde and retrograde axonal transport from slow transport and state what each carries.

Do one of the two. Head your paper with the number.

11-1

Draw a long neuron with the cell body on the left and the terminal on the right, and run a microtubule track down the axon. On the track draw fast anterograde transport with kinesin and fast retrograde transport with dynein, arrows in opposite directions, with the cargo written beside each. One line: the cell body makes everything, so say what the retrograde direction is even for.

11-2

Draw the same axon and add slow transport as a third, much slower stream with its cargo written beside it. Draw a scale bar showing the real speed difference, hundreds of mm per day for fast transport against a few mm per day for slow. Then draw a virus entering at the terminal and mark the route it takes to the cell body. One line: say which transport system it hijacked and why that is bad news for the neuron.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Neurons and glia. Conduction speeds, transport speeds and regrowth rates are typical textbook values and are approximate. All figures are original drawings for this course.